

DIARRRHEAL DISEASE

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Outline

- Definition
- Epidemiology
- Etiology
- Classification
- Diagnosis
- Management

Diarrheal Disease

- **Diarrhea**; it's passage of 3 or more loose or watery stools with in a 24-hour. or stool volume of more than 20 grams/kg/day in infants and toddlers (<10 kg) or more than 250 grams/day in older children or teenagers.
- **Acute gastroenteritis (AGE)**; comprises infectious cases of diarrhea.
- **Watery diarrhea**; refers to passage of ≥ 3 watery stools per day with no visible blood.
- **Invasive (bloody) diarrhea, or dysentery**; refers to the passage of ≥ 3 stools per day mixed with visible blood

Epidemiology

- Globally, there are nearly 1.7 billion cases of childhood diarrheal disease every year.
- Each year diarrhea kills around 443 832 children under 5 and an additional 50 851 children aged 5 to 9 years.
- Approximately 88% of those deaths occurred in sub-Saharan Africa and South Asia (71% and 17%, respectively).
- Diarrheal disease is the third leading cause of death in under 5yr children.
- It is both preventable and treatable.
- Also a leading cause of malnutrition in children under 5 years old.

Classification

Based on duration	Acute diarrhea (lasting <7day)
	Prolonged diarrhea (7-14day)
	Persistent diarrhea (≥ 14 days)
	Chronic diarrhea (>4week)
Based on clinical manifestation	Acute watery diarrhea
	Invasive (bloody diarrhea)
Based on underlying mechanism	Secretory diarrhea
	Osmotic diarrhea
	Inflammatory diarrhea
	Motility related
	Exudative diarrhea
Based on Etiology	Infectious (Viral, Bacterial, Parasitic)
	Non-infectious (Drug, Dietary, Anatomical, Inflammatory)

Etiology of Acute diarrhea

VIRAL	BACTERIAL	PARASITIC
➤ Rotavirus ➤ Caliciviruses (including noroviruses and sapoviruses) ➤ adenovirus, 40 and 41 ➤ SARS-CoV-2 ➤ Astroviruses	➤ Nontyphoidal Salmonella (NTS), ➤ Shigella, ➤ Campylobacter, ➤ Yersinia ➤ E. coli {(STEC), (EHEC), (ETEC), (EPEC), (EAEC), (EIEC)}. ➤ Vibrio cholerae (O1 and O139) ➤ Clostridioides difficile ➤ Bacillus cereus, ➤ Clostridium perfringens, ➤ Staphylococcus aureus	➤ Cryptosporidium ➤ Cyclospora <u>cayetanensis</u> ➤ Entamoeba histolytica ➤ Giardia <u>duodenalis</u>

Major etiologies of childhood diarrhea in resource-limited countries

Syndrome	Etiologic agents	Features
Acute watery diarrhea Watery stools; may contain mucous. Fever may be present.	Rotavirus	Leading cause of gastroenteritis in children younger than two years.
	Enterotoxigenic <i>Escherichia coli</i> (ETEC)	Leading cause of gastroenteritis in older children and adults.
	<i>Vibrio cholerae</i> O1 and O139	Associated with endemic and epidemic disease. Vomiting and voluminous "rice-water diarrhea" in severe cases.
	Cryptosporidium	Common in infants (younger than one year) even in the absence of HIV; infrequently seen in older children.
	Norovirus	Abrupt onset of vomiting and diarrhea with low grade fever.
Invasive diarrhea Gross blood in stool. Often associated with fever, vomiting, abdominal pain.	Shigella spp.	Leading cause of invasive diarrhea. <i>S. dysenteriae</i> serotype I produces Shiga-toxin and is associated with epidemics of severe disease. Complications include toxic megacolon, rectal prolapse, intestinal perforation, seizures, encephalopathy and sepsis.
	Nontyphoidal <i>Salmonella enterica</i>	Several serotypes cause gastroenteritis. Infants, elderly, and immunocompromised at increased risk for disseminated infection.
	Campylobacter spp.	Predominantly <i>C. jejuni</i> and <i>C. coli</i> . May mimic appendicitis. Complications include Guillain-Barré syndrome.
	Enteroinvasive <i>Escherichia coli</i> (EIEC)	EIEC are closely related to Shigella and cause a syndrome essentially identical to shigellosis.
	Enterohemorrhagic <i>Escherichia coli</i> (EHEC)	EHEC produce Shiga toxin identical to that produced by <i>S. dysenteriae</i> serotype I, associated with increased risk of hemolytic uremic syndrome.
	<i>Entamoeba histolytica</i>	<i>E. histolytica</i> is a protozoal organism which causes intestinal infection which may be indistinguishable from Shigella and other bacteria. Rare complications include extraintestinal infections, most commonly hepatic abscess.
	Adenovirus types 40/41	Also cause watery diarrhea.

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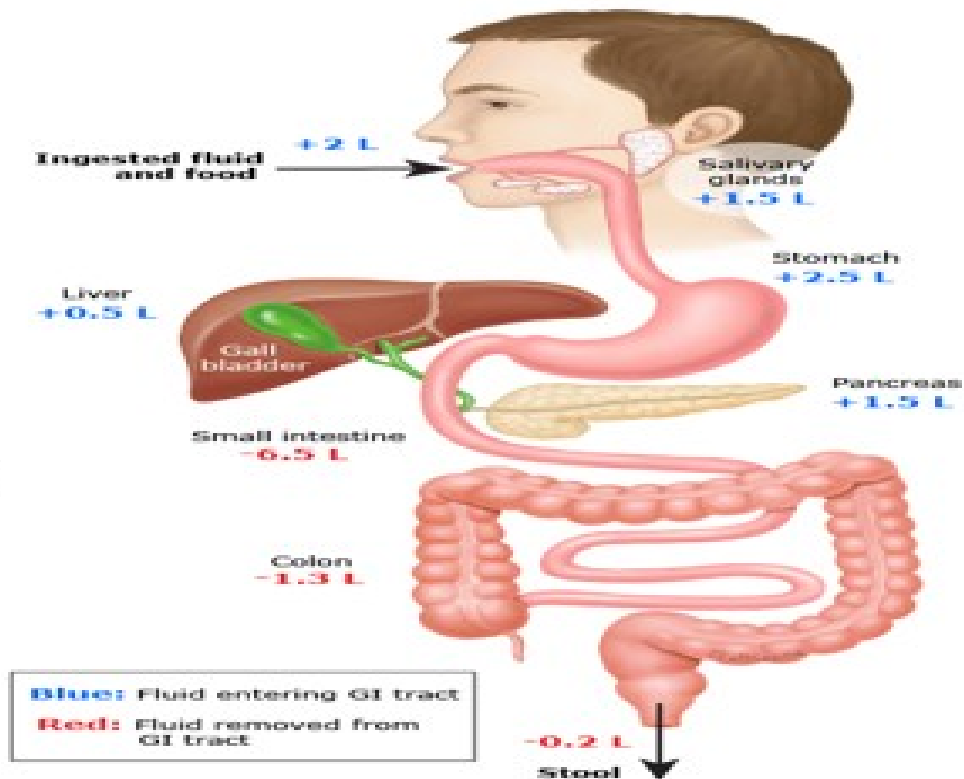
- ❖ Most episodes of moderate to severe diarrhea were attributed to four pathogens:
 - Rotavirus,
 - Cryptosporidium,
 - Shigella, and
 - ETEC producing heat-stable toxin (ST)
- ❖ Three agents associated with most deaths among children <5 years ;
 - ✓ Rotavirus (29%),
 - ✓ Cryptosporidium (12%), and
 - ✓ Shigella (11%)

Risk factors

HOST FACTOR	ENVIRONMENTAL FACTOR
❖ In appropriate breast-feeding practice ○ No EBF for first 6mo, ○ Early interruption, bottle feeding ❖ Unsanitary food preparation ❖ Lack of immunization ❖ Young age ❖ Measles infection ❖ Malnutrition ❖ Immunocompromised ❖ Vit A and zinc deficiency	➤ Seasonality ➤ In adequate food intake ➤ Poverty and poor living condition ➤ Poor domestic and environmental sanitation

Pathogenesis of Diarrhea

Fluid movement in the human gastrointestinal tract



Schematic diagram of the GI tract showing daily volumes (in liters) of fluid secretion and absorption.

□ Mechanism of diarrhea

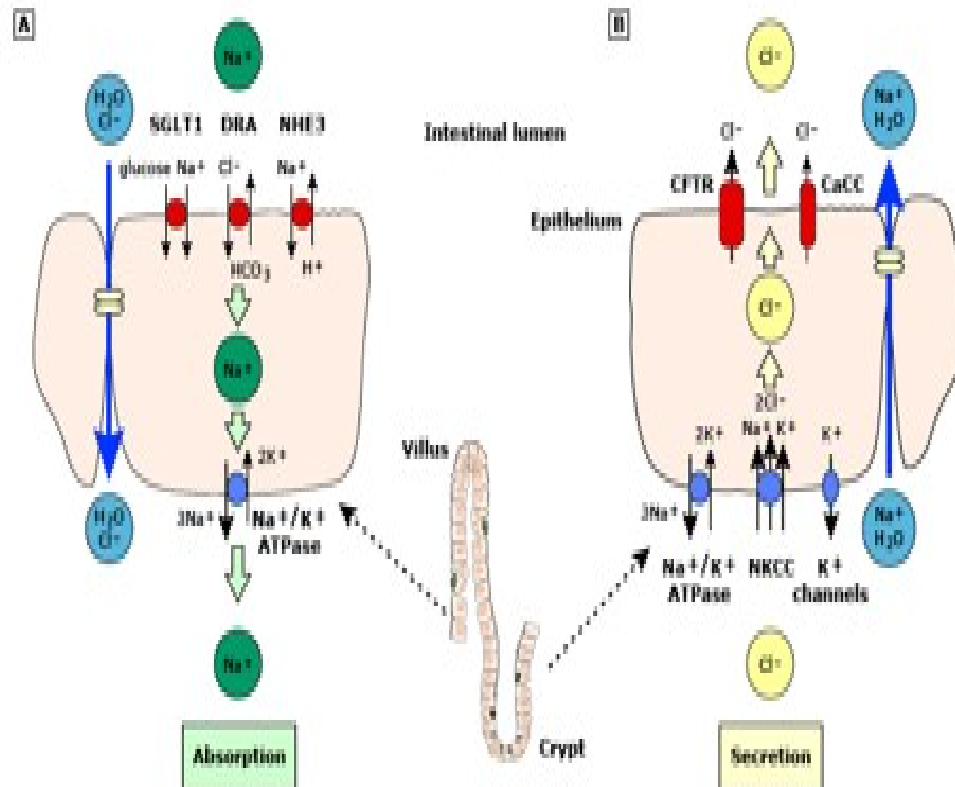
- Secretory diarrhea
- Osmotic diarrhea
- Motility-related
- Inflammation-related

Secretory diarrhea

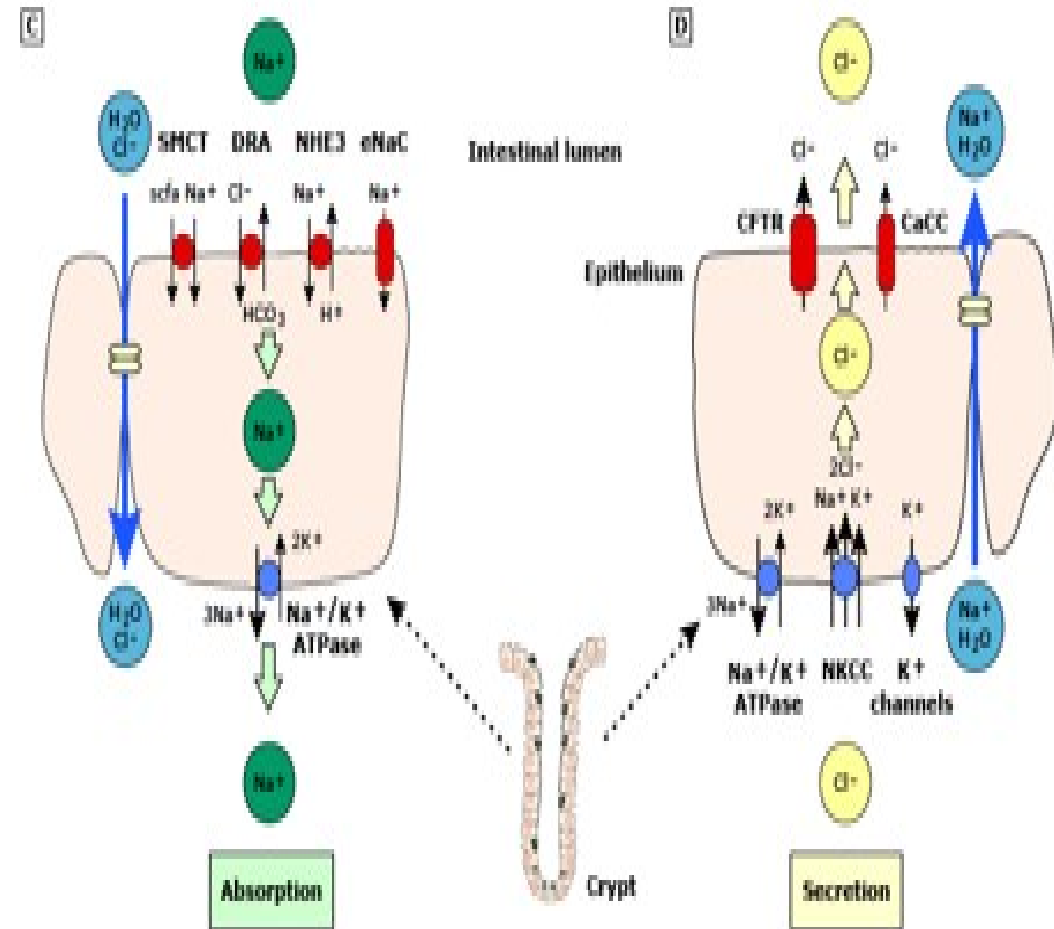
- Characterized by active electrolyte and water fluxes toward the intestinal lumen
- Result from the effect of one of the following mechanism
 - ✓ Inhibition of neutral NaCl absorption in villous enterocytes
 - ✓ Increase in electrogenic chloride secretion in secretory crypt cells as a result of the opening of the cystic fibrosis transmembrane regulator (CFTR) chloride channel
 - ✓ Na-k -2cl co-transporter for the electroneutral chloride entrance into the enterocyte;
 - ✓ Na-k pump, which decreases the intracellular Na^+ concentration, determining the driving gradient for further Na^+ influx; and
 - ✓ K^+ selective channel, that enables K^+ , once it has entered the cell together with Na^+ , to return to the extracellular fluid

Intestinal transport mechanisms

Small intestine



Colon



Osmotic diarrhea

❑ Caused by nonabsorbed nutrients in the intestinal lumen as a result of one or more of the following mechanisms:

- (1) Intestinal damage (e.G., Enteric infection);
- (2) Reduced absorptive surface area (e.G., Active celiac disease);
- (3) Defective digestive enzyme or nutrient carrier (e.G., Lactase deficiency);
- (4) Decreased intestinal transit time (e.G., Functional diarrhea); and
- (5) Nutrient overload, exceeding the digestive capacity (e.G., Overfeeding, sorbitol in fruit juice)

Clinical manifestation

- GI;- Nausea, vomiting, diarrhea, abdominal pain
- Systemic;- fever, loss of appetite, myalgia, UTI, meningitis, sign and symptom of dehydration and shock,
- Immune-mediated manifestation
 - GBS
 - HUS
 - Reactive arthritis

Complication

- Dehydration and shock
- Acute renal failure
- Sepsis /DIC
- Metabolic acidosis
- Convulsion and coma
- Paralytic ileus
- Malnutrition
- Persistent diarrhea

Intestinal and Extraintestinal Complications of Enteric Infections

INTESTINAL COMPLICATIONS	EXTRAINTESTINAL COMPLICATIONS	POSTINFECTIOUS COMPLICATIONS
• Persistent diarrhea • Recurrent diarrhea (usually immunocompromised persons) • Postinfectious irritable bowel syndrome • Protein-losing enteropathy • Toxic megacolon • Intestinal perforation • Rectal prolapse • Pseudomembranous colitis • Appendicitis • Intussusception	• Dehydration, metabolic abnormalities, malnutrition, micronutrient deficiency • Systemic invasion with bacteremia/parasitemia ± distant foci • Local invasion Mesenteric adenitis Cholecystitis Campylobacter spp. • Local noninvasive spread Vulvovaginitis Urinary tract infection • Seizures, encephalopathy • STEC Leukemoid reaction, Pharyngitis, adenopathy, rash	• Reactive arthritis • Guillain-Barré syndrome • Hemolytic uremic syndrome • Glomerulonephritis, • Myocarditis, • Pericarditis • Nephropathy

Evaluation

➤ History

- Vomiting and diarrhea (type ,frequency, volume ,duration, onset)
- Associated symptom(fever, change in mentation, seizure, abdominal pain, and other extra-intestinal manifestation)
- Risk factor

➤ Physical exam

- General appearance
- V/S
- Sign of dehydration and shock
- Associated systemic illness

➤ Diagnostic work up (stool exam, stool culture, serum electrolyte, RFT,CBC)

Classification of dehydration

WHO guidelines for assessment of dehydration

No dehydration (<5%)	Some dehydration (5 to 10%)	Severe dehydration (>10%)
	Two or more of the following signs:	Two or more of the following signs:
▪ Well, alert	▪ Restlessness, irritability	▪ Lethargy or unconsciousness
▪ Normal eyes	▪ Sunken eyes	▪ Sunken eyes
▪ Drinks normally, not thirsty	▪ Thirsty, drinks eagerly	▪ Unable to drink or drinks poorly
▪ Skin pinch goes back quickly	▪ Skin pinch goes back slowly	▪ Skin pinch goes back very slowly (≥ 2 seconds)
▪ Estimated fluid deficit: <50 mL/kg	▪ Estimated fluid deficit: 50 to 100 mL/kg	▪ Estimated fluid deficit: >100 mL/kg

Management of Acute diarrhea in children

- Assess for nutritional status
- Correcting fluid and electrolyte losses,
- Administering appropriate nutrient, and micronutrient
- Managing associated comorbid conditions.
- Antibiotic therapy

Fluid and electrolytes

- ❑ Rehydration therapy is critical for management of diarrhea.
- ❑ The goal of replacement therapy is to replenish deficits in water and electrolytes lost.
- ❑ **Phases of treatment** – Fluid management consists of two phases
 - ✓ **Replacement** – Continues until all signs and symptoms of volume depletion have resolved and the patient has urinated;
 - ✓ **Maintenance** – The maintenance phase counters ongoing losses of water and electrolytes; this phase continues until diarrhea and other signs and symptoms of illness have resolved.

No signs of dehydration (Plan A)

- Can be managed as out patient
- ORS used to replace ongoing loss
- Supplemental home made fluid
- Advice on danger sign (progression of DHN)
- In general, children ≤ 2 years old should receive approximately 50 to 100 mL of ORS for each episode of diarrhea or vomiting; children > 2 years old should receive 100 to 200 mL of ORS for each episode

Some dehydration (Plan B)

- Patient managed at rehydration corner
- Replacement fluid with ORS 75ml/kg/4hr then re assess for response and reclassify
- Replace ongoing loss according to the age

Severe dehydration (Plan C)

- Requires treatment with IV fluids (RL or NS).

Age	1 st give 30ml/kg in	Then give 70ml/kg in
< 12 months	1 hour	5 hours
≥ 12 months	30 minute	Two & half hours

- Then reassess if improved manage for some DHN
- Replace ongoing loss
- Start antibiotics if indicated

Vitamins and minerals

- **Zinc** – for children 6 months to 5 years of age at a dose of 10 mg/day for 14 days. It reduce the duration of a diarrhea episode by 25% and are associated with a 30% reduction in stool volume
- **Vitamin A** –is warranted for patients with acute diarrhea who have signs and symptoms of vitamin A deficiency (such as xerophthalmia, keratitis, keratoconjunctivitis, corneal ulceration, or Bitot spots),measles ,malnutrition.

Antibiotics

- Cholera suspected
- Patients with invasive (bloody) diarrhea
- Patients with persistent symptoms
- Patients with amebiasis, giardiasis
- Patients with severe acute malnutrition
- Patients with tachypnea as well as cough and/or chest retractions

Prevention

- ❖ **Key measures to prevent diarrhea include:**
 - Access to safe drinking-water
 - Use of improved sanitation
 - Hand washing with soap
 - Exclusive breastfeeding for the first 6 months of life and appropriate nutritional supplement
 - Good personal and food hygiene
 - Health education about how infections spread
 - Immunization against rotavirus, measles cholera

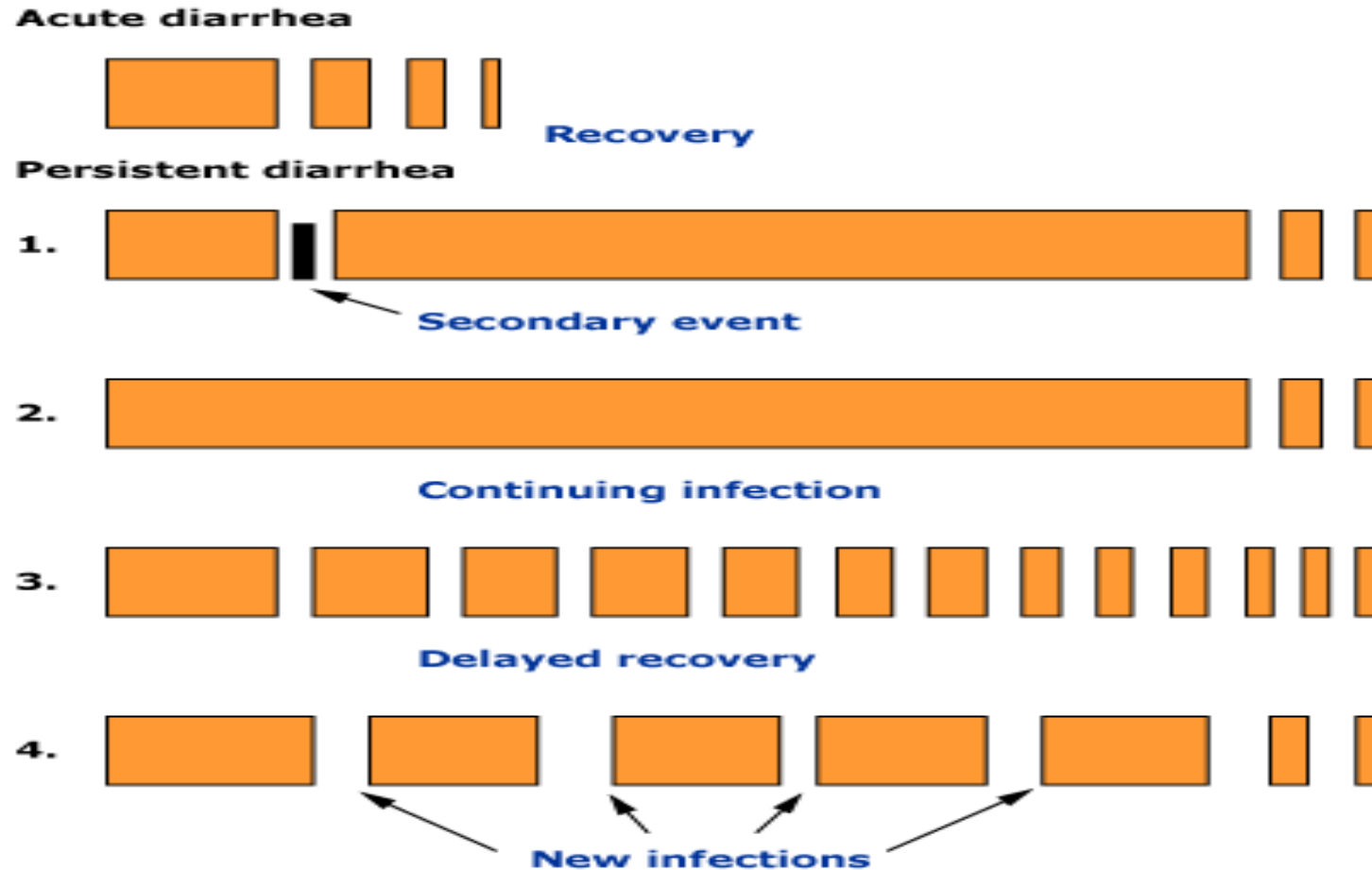
Persistent Diarrhea

- Defined as diarrhea lasting 14 days or longer.
- In resource-limited settings, it is often associated with
- Chronic enteropathy, with
 - Impaired mucosal healing and
 - Diminished digestive and absorptive capacity, resulting in malabsorption or maldigestion

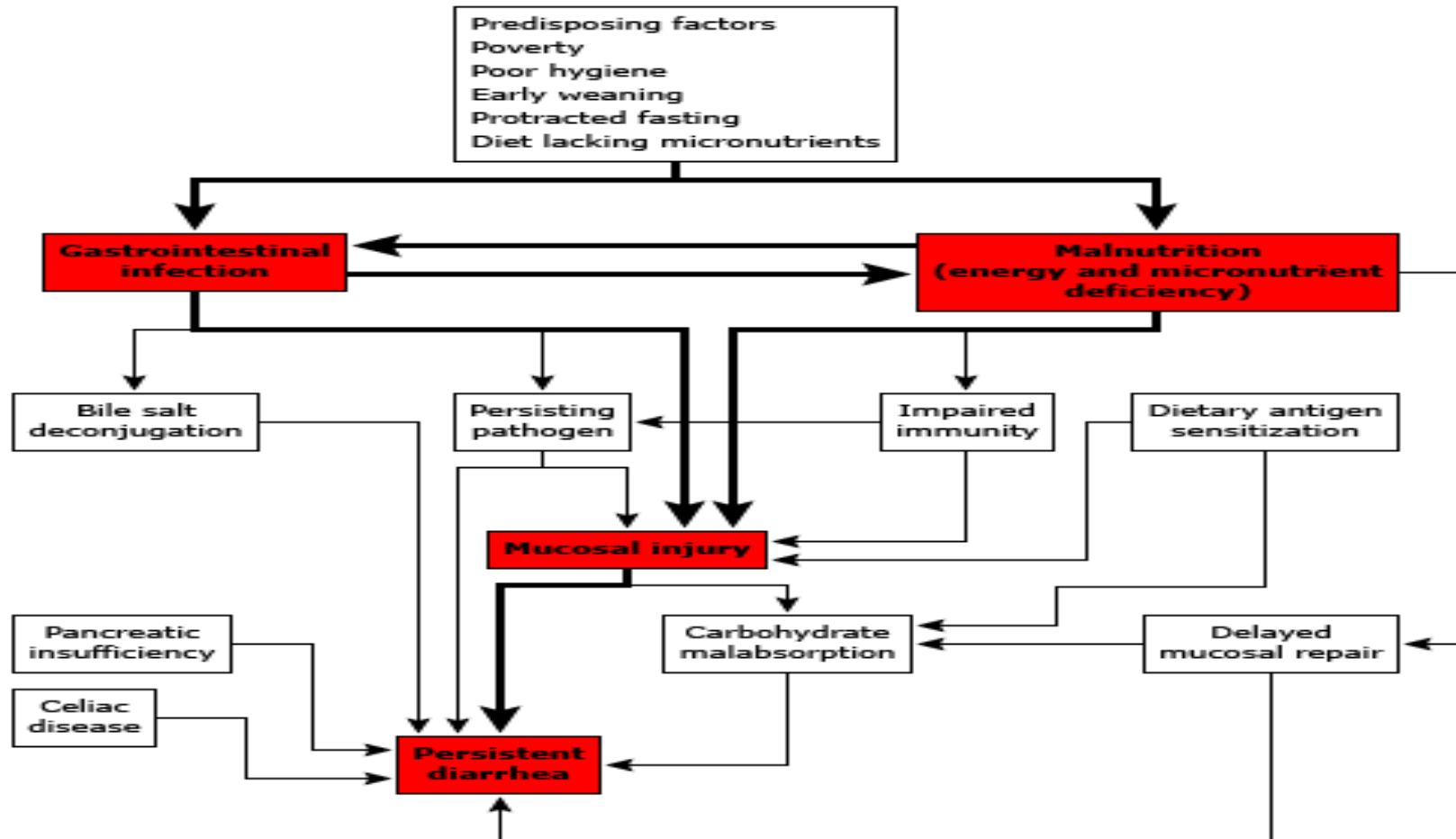
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- In resource-limited settings, persistent diarrhea usually follows an acute episode and typically is associated with serial enteric infections without sufficient time to recover between episodes.
- It is most common in infants and toddlers but can also occur in older children
- The extent to which diarrhea episodes become persistent is highly variable.
- Overall, fewer than 10 percent of episodes of diarrhea persist for two or more weeks,
- **Risk depends** on many factors, including the initial enteropathogen, nutritional status, geographical location, and socioeconomic condition

Explanatory mechanisms for persistent diarrhea



Causes of persistent diarrhea in children in resource-limited countries



Diagnostic approach

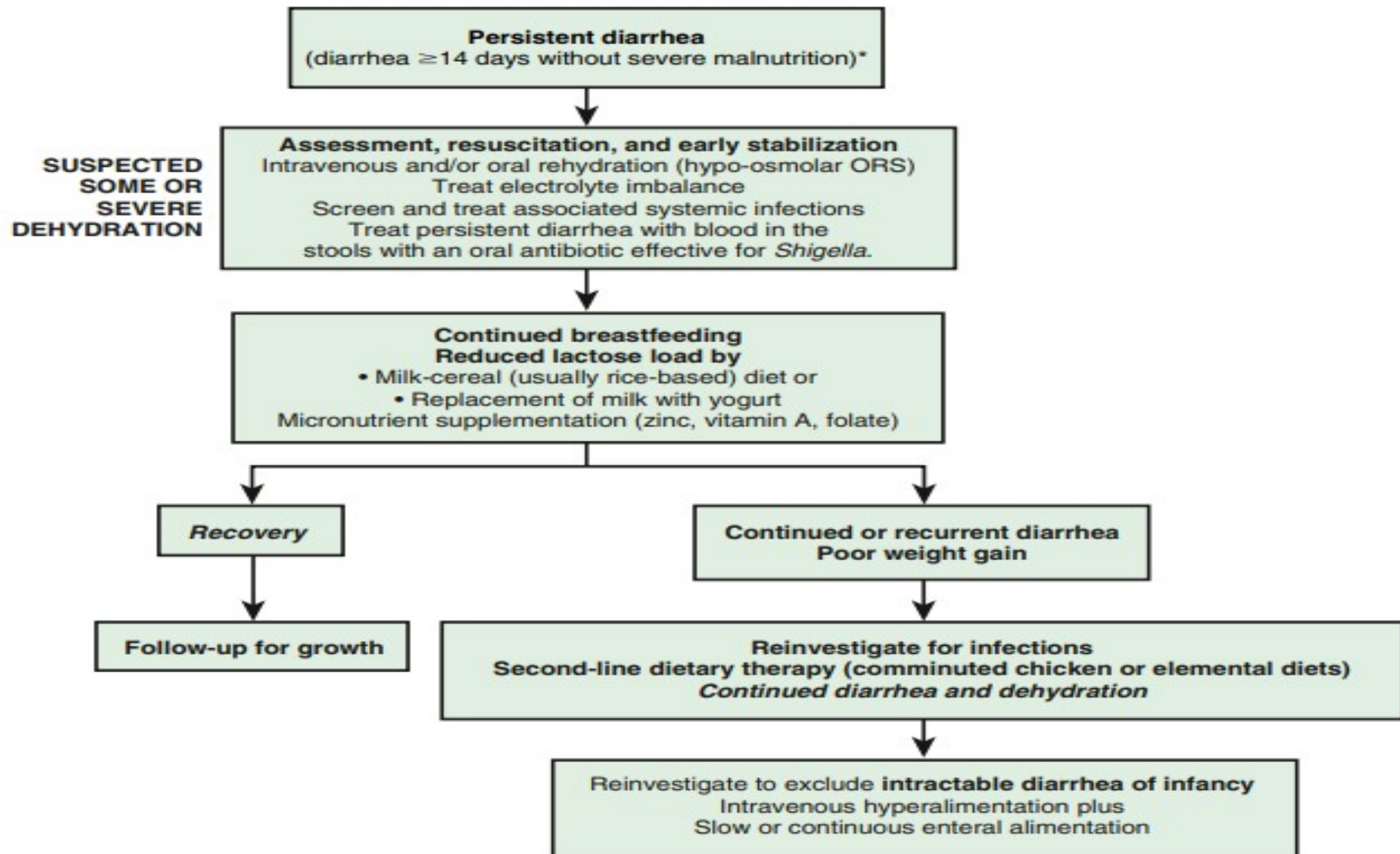
- Assess for complications and comorbid conditions
 - Dehydration
 - Malnutrition
 - Infections

*And if diarrhea
14 days or more*

• Dehydration present	Severe persistent diarrhea	• Treat dehydration before referral unless the child has another severe classification. • Refer to hospital.
• No dehydration	Persistent diarrhea	• Advise the mother on feeding a child who has PERSISTENT DIARRHEA. • Give multivitamin, mineral supplement for two weeks • Advise mother when to return immediately • Follow-up in 5 days.

Treatment

- Treatment of persistent diarrhea includes
 - ✓ Supportive measures,
 - ✓ Nutritional rehabilitation,
 - ✓ Drug therapy if indicated
 - ✓ Micronutrients and vitamins



Chronic Diarrhea

- Passage of 3 or more loose or watery stool that lasts for 4 weeks or more
- In resource-limited settings, chronic diarrhea is common and associated with significant mortality and morbidity.
- The prevalence of chronic infectious diarrhea (CID) in childhood is unknown and significantly varies according to the geographical and clinical setting.
- In low-income countries, 12–35% of children with acute diarrhea undergo a prolonged course, and 5–7% of cases last >14 days

Causes of chronic diarrhea in children

Physiologic
Lactase nonpersistence (hypolactasia)*
Infection-related
Postinfectious malabsorptive diarrhea* - Lactose or other specific carbohydrate intolerance without residual inflammation* - Generalized malabsorption with residual inflammation
Persistent or recurrent infection - Primary immunodeficiency (SCID, IPEX, CVID)[¶] - Secondary immunodeficiency (eg, transplant patient on immunosuppressive drugs, HIV infection, or child with malnutrition)[¶] - No underlying immunodeficiency – Caused by organisms that are difficult to eradicate, such as <i>C. difficile</i> or <i>S. typhi</i>
Drug-induced
Antibiotic-associated diarrhea*
Non-antibiotic-associated - Without mucosal injury (eg, laxatives, SSRIs, orlistat) - With mucosal injury (eg, 5-FU, NSAIDs, olmesartan)

Fat malabsorption
Pancreatic exocrine insufficiency - Cystic fibrosis* - Shwachman-Diamond syndrome
Bile acid insufficiency - Resection of the terminal ileum - Bile salt deconjugation (eg, in small intestine bacterial overgrowth)
Mucosal malabsorption - Short bowel syndrome - Severe and diffuse small intestinal Crohn disease
Neuroendocrine tumors and related disorders - Neurofibromatosis (associated with several types of neuroendocrine tumors) - Neuroblastoma and ganglioneuroblastoma (eg, VIPoma) - Multiple endocrine neoplasia type 2b

Functional diarrhea
Functional diarrhea in young children*^Δ Often associated with excessive sugar or carbohydrate intake
Irritable bowel syndrome (diarrhea-predominant)
Immune-mediated
Common causes - Celiac disease* - Inflammatory bowel disease* - Food protein-induced allergic proctocolitis (eg, due to cow's milk)*[◇]
Uncommon causes - Food protein-induced enterocolitis - Eosinophilic gastroenteritis - Microscopic lymphocytic or collagenous colitis - Autoimmune enteropathy - Primary immunodeficiencies (refer to above) - Polyglandular autoimmune syndrome - Cutaneous or systemic mastocytosis - Graft-versus-host disease

Congenital anomalies associated with bowel obstruction
- Chronic intestinal pseudo-obstruction - Hirschsprung disease, with enterocolitis - Malrotation with intermittent volvulus

Other mechanisms
- Small intestine bacterial overgrowth - Bile acid malabsorption - Protein-losing enteropathy/gastropathy - Factitious diarrhea - Congenital diarrheas and enteropathies (eg, sucrase-isomaltase deficiency) - Vasculitis

Evaluation of patients

- Chronic diarrhea is typically evaluated in an outpatient setting.
- Selected patients with warning signs may require more extensive or urgent evaluation, sometimes in the inpatient setting.
- Warning signs
 - ✓ Fever
 - ✓ Gross blood in stool
 - ✓ Growth faltering or weight loss
 - ✓ Nausea, vomiting
 - ✓ Severe abdominal distension or tenderness
 - ✓ Hepatosplenomegaly or mass

Key elements of the history for a child age >6 months with chronic diarrhea

Historical clues	Implications
Pattern of initial onset	
Abrupt	Suggests infectious/postinfectious or drug- or dietary-induced diarrhea.
Gradual	Can be seen in many disorders, including IBD and functional diarrhea.
Potential triggers	
Dietary changes	
Initial onset after a change from breast milk to formula or from formula to unmodified cow's milk	Possible food protein-induced allergic proctocolitis or enterocolitis (usually related to cow's milk).
Initial onset after an increase in dietary fructose, sucrose, or lactose or poorly absorbed carbohydrates (eg, sorbitol, xylitol)	Multiple possible mechanisms: ▪ High intake may cause osmotic diarrhea (dose-related) or exacerbate functional diarrhea. ▪ Temporary disaccharidase deficiency (eg, postinfectious). ▪ Permanent disaccharidase deficiency. Physiologic lactase nonpersistence (hypolactasia) is common; other disaccharidase deficiencies are rare.
Onset coincided with introduction of gluten (eg, in wheat, rye, or barley)	Possible celiac disease. In severe cases, onset of symptoms may occur within weeks to months after gluten is introduced (or increased) but may occur at any time thereafter.
Infectious exposures	
Onset after acute diarrheal illness (in index patient or close contacts)	Infectious or postinfectious diarrhea. Postinfectious diarrhea often varies with diet (eg, exacerbated by lactose).
Recent travel, particularly to equatorial/tropical countries with endemic enteric pathogens	Potential prolonged enteric infection or postinfectious diarrhea.
Recent visit to campgrounds; water supply from well	Consideration of parasitic exposure (eg, <i>G. lamblia</i>).
Drugs	
Onset associated with the use of certain drugs (eg, antibiotics, SSRIs)	Possible drug-related side-effect*.
Physical symptoms	
Bloating symptoms	Suggests abnormality of carbohydrate assimilation or small intestine bacterial overgrowth.
Nausea/vomiting	Suggests intestinal inflammation (including FPIES) or impaired motility.

Diarrhea pattern	
Postprandial episodes	Suggests food intolerance or functional diarrhea (eg, IBS-D).
Nocturnal episodes	Suggests an organic disorder with a component of abnormal electrolyte absorption/secretion. One exception is that young children with functional diarrhea sometimes have very low-volume nocturnal episodes of diarrhea.
Stool appearance	
Blood and/or mucus	Food protein-induced proctocolitis (eg, due to cow's milk), IBD, prolonged enteric infection (eg, <i>C. difficile</i>).
Light tan or white	Absence of bile, suggesting biliary obstruction. Occasionally found in celiac disease.
Family history	
Family history of autoimmunity, IBD, or other significant systemic or gastrointestinal disorders	Increases the likelihood of heritable diseases, including celiac disease and IBD (both relatively common) or any other polygenic and monogenic disorders (rare).
Signs/symptoms suggesting underlying medical illnesses	
History of recurrent, persistent, or opportunistic infections or immunodeficiency diagnosis	Possible immunodeficiency, immune-mediated diarrhea, or graft-versus-host disease.
Growth faltering or weight loss	Malabsorption, pancreatic exocrine insufficiency, anorexia nervosa.
Intravenous drug use or other risk factors	HIV disease.
History of chronic constipation and/or fecal incontinence	Possible retentive (constipation-associated) fecal incontinence (misinterpreted as diarrhea by the patient/family).

Key examination findings for a child age >6 months with chronic diarrhea

Physical examination findings	Implications
Aphthous stomatitis	IBD and celiac disease
Stomatitis/glossitis/angular cheilitis	B-complex vitamin deficiency secondary to malabsorption
Diaper/perianal rash	- Acidic stools – Carbohydrate malabsorption - Acrodermatitis enteropathica – Zinc deficiency
Cutaneous rashes	- Dermatitis herpetiformis – Celiac disease - Erythema nodosum – IBD - Pyoderma gangrenosum – IBD
Loss of subcutaneous fat and muscle loss	Chronic malnutrition
Perianal fistula	Crohn disease
Abdominal tenderness	Inflamed organs or peritoneum (IBD, vasculitis, and others)
Abdominal distension	- Excessive gas – Carbohydrate fermentation, malabsorption, or SIBO - Ascites – Suggest liver or PLE
Abdominal distension	- Excessive gas – Carbohydrate fermentation, malabsorption, or SIBO - Ascites – Suggest liver or PLE
Peripheral edema	Suggests PLE
Koilonychia (spoon nails)	Iron deficiency secondary to malabsorption (also may be idiopathic)
Xeroderma, dry skin	Essential fatty acid deficiency secondary to malabsorption

Table 388.4 Noninvasive Tests for Intestinal Digestive–Absorptive Function and Inflammation

TEST	NORMAL VALUES	IMPLICATION
α_1 -Antitrypsin concentration	<0.9 mg/g	Increased intestinal permeability/protein loss
Steatocrit	<2.5%-fold (older than 2 yr) increase over age-related values (younger than 2 yr)	Fat malabsorption
Fecal-reducing substances	Absent	Carbohydrate malabsorption
Elastase concentration	>200 μ g/g	Pancreatic function
Chymotrypsin concentration	>7.5 units/g >375 units/24 hr	Pancreatic function
Fecal occult blood	Absent	Blood loss in the stools/inflammation
Fecal calprotectin concentration	<100 μ g/g (in children upto 4 yr of age) <50 μ g/g (older than 4 yr)	Intestinal inflammation
Fecal leukocytes	<5 per microscopic field	Colonic inflammation
Fecal lactoferrin	Absent	Inflammation
Nitric oxide in rectal dialysate	<5 μ M of $\text{NO}_2^-/\text{NO}_3^-$	Rectal inflammation
Dual sugar (cellobiose/mannitol) absorption test	Urine excretion ratio: 0.010 ± 0.018	Increased intestinal permeability
Xylose oral load	25 mg/dL	Reduced intestinal surface

Treatment

❖ Treatment includes

- General supportive measures,
- Nutritional rehabilitation,
- Elimination diet
- Medications

Reference

- Nelson 22nd edition
- Rudolf pediatrics text book
- AAP 2024
- WHO Diarrheal guideline
- UpToDate 2024



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